Overview

Presentation Confidence Detector

Project Blueprint

What is the Presentation Confidence Detector?

The Problem We're Solving

Picture this real scenario:

- A professional is preparing for a **big client presentation** tomorrow
- They practice alone in their room, talking to the wall
- They THINK they sound confident... but do they?
- They have no way to know if they're saying "um" too much, avoiding eye contact, or speaking too fast
- They show up tomorrow, deliver the presentation, and the client says: "You didn't seem very sure about this"

Can a person accurately judge their own confidence?

No. Research shows that people are terrible at self-assessment. Nervous speakers often think they did fine. Over-confident speakers don't realize they're being arrogant. Without external feedback, you're guessing.

Hiring a coach costs $200-500/hour. Recording yourself and watching the video takes 2x the time and you still miss things. Most people just... wing it.

The Solution: Presentation Confidence Detector

A system that watches you present and gives you the feedback a professional coach would — but instantly, freely, and privately.

Think of it like this:

> ***Without the Detector:*** *You practice alone, guess how you did, hope for the best*
>
> ***With the Detector:*** *AI watches your face, listens to your words, measures your voice — and tells you exactly what to fix*

The system doesn't judge you. It doesn't grade you. It measures specific signals:

"You said 'um' 14 times in 3 minutes. Your eye contact dropped to 40% in the second half. You spoke at 190 words per minute — that's too fast. When you talked about pricing, your voice got shaky."

That's actionable feedback. That's what a $300/hour coach would tell you.

How It Works (The Pipeline)

```
Step 1: USER starts a practice session
     |
     v
Step 2: CAMERA captures your face in real-time
     |
     v
Step 3: MICROPHONE captures your voice
```

```
     |
     v
Step 4: FOUR DETECTION ENGINES run simultaneously
     |
     ├── Face Engine: expressions, eye contact, blink rate
     ├── Speech Engine: filler words, hedging, repetitions
     ├── Voice Engine: pitch, volume, shakiness, pace
     └── AI Engine: deep analysis combining everything (periodic)
     |
     v
Step 5: SCORING ALGORITHM combines all signals → single confidence score (0-100)
     |
     v
Step 6: LIVE FEEDBACK shows on dashboard while presenting
     |
     v
Step 7: SESSION REPORT gives detailed breakdown after stopping
```

Step-by-Step Breakdown:

Step 1 - Session Start:

User enters a topic ("Q3 Revenue Report") and clicks Start. Camera and microphone activate.

Step 2 & 3 - Capture:

Camera sends 30 images per second (video frames). Microphone sends continuous audio. Both feed into the detection engines.

Step 4 - Detection (4 engines in parallel):

Engine	What It Analyzes	What It Outputs	How Often
Face Engine	468 face landmark points	Expression (tense/neutral/confident), eye contact %, blink rate	Every frame (30x/sec)
Speech Engine	Transcribed text from STT	Filler count, hedging score, repetitions, pace (WPM)	Every final sentence
Voice Engine	Raw audio signal	Volume, pitch variation, shakiness, silence ratio	Continuous
AI Engine	Webcam frame + recent transcript	Deep written analysis, coaching suggestions	Every 30 seconds

Step 5 - Scoring:

All four engine outputs combine into one number: **Confidence Score (0-100)**. Weighted so that no single signal dominates.

Step 6 - Live Feedback:

Dashboard shows the score updating in real-time. Coaching nudges appear: "Slow down", "Make eye contact", "You're doing great!"

Step 7 - Report:

After stopping, a detailed report breaks down everything: score timeline, filler chart, eye contact graph, AI coaching notes, top 3 things to improve.

Core Capabilities

Capability	What It Detects	Technology	Example
Expression Reading	Nervousness, tension, genuine vs forced smile	MediaPipe Face Mesh (468 landmarks)	"Lip tension detected — you look stressed"
Eye Contact	Looking at camera vs looking away	Iris tracking + head direction	"Eye contact dropped to 35% during pricing section"
Filler Words	"um", "uh", "like", "you know"	Speech-to-Text + pattern matching	"14 fillers in 3 minutes = 4.7 per minute (high)"
Hedging Language	"I think maybe", "sort of", "I'm not sure"	Text pattern matching	"5 hedging phrases when discussing data — sounds unsure"
Speaking Pace	Words per minute, speeding up/slowing down	Word count over time	"Average 175 WPM — optimal is 130-160"
Voice Steadiness	Shaky voice, volume drops, monotone	Audio frequency analysis	"Voice pitch wavered during conclusion — nervousness"
AI Coaching	Overall assessment combining all signals	Claude API (vision + text)	"Strong content but delivery needs work — here's what to practice"

The Golden Rule: Measure, Don't Judge

```
+--------------------------------------------------+
|                                                    |
|   The system MEASURES, it does not JUDGE           |
|                                                    |
|   - It counts fillers   → doesn't call you stupid |
|   - It tracks eye contact → doesn't say you're shy|
|   - It scores confidence → doesn't decide you're  |
|     bad at presenting                              |
|                                                    |
|   The user decides what to work on.                |
|   The system just shows the data.                  |
|                                                    |
+--------------------------------------------------+
```

Why this matters:

1. **Everyone starts somewhere** — a low score isn't failure, it's a baseline

2. **Context matters** — looking away might mean thinking, not nervousness
3. **Culture matters** — some cultures avoid eye contact out of respect
4. **The user is in control** — they choose what feedback to act on

---

How Does This Compare to Existing Products?

There are commercial products in this space. Here's an honest comparison:

Feature	Yoodli	Poised	Orai	Ours
What it does	AI speech coach — records you, analyzes filler words, pacing, eye contact	Real-time overlay during Zoom/Teams calls — coaches during meetings	Mobile app for speech practice — grades your delivery	Browser-based confidence detector — face + speech + voice scoring
Real-time feedback	No — analyze after recording	Yes — during live meetings	No — post-practice review	Yes — live dashboard while presenting
Offline capable	No — cloud only	No — cloud only	No — cloud only	Yes — face and voice analysis run 100% in the browser. STT depends on engine choice
Open source	No	No	No	Yes — you own everything, can modify anything
Customizable weights	No — their algorithm, their rules	No	No	Yes — you decide Face 0.40 / Speech 0.35 / Voice 0.25 or whatever you want
Privacy	Video uploaded to their servers	Audio processed on their servers	Audio processed on their servers	Video and audio never leave your device (except optional Claude API calls)
Cost	Free tier + $8-25/month	Free tier + $8-16/month	Free tier + $10/month	Free forever — you run it yourself
Face analysis	Basic (eye contact only)	None (audio only)	None (audio only)	Full — 468 landmarks, expressions, eye contact, blink stress

---

WHY Build This Instead of Using Yoodli?

Four reasons:

1. Control. Commercial products give you THEIR algorithm with THEIR thresholds. If Yoodli says 130 WPM is "too slow," you can't change it. With ours, you set every threshold because you wrote the code.

2. Privacy. Every commercial option uploads your video or audio to their cloud. If you're practicing a confidential investor pitch or a medical presentation, that's a problem. Our system processes everything locally in the browser.

3. Learning. Using Yoodli teaches you nothing about how AI works. Building this system teaches you real-time media capture, computer vision, NLP, audio analysis, and system design — skills that transfer to dozens of other projects.

4. Customization. Want to add industry-specific filler words? ("basically" in tech, "synergize" in consulting). Want to weight eye contact higher for sales presentations and lower for technical demos? Commercial products can't do that. Yours can.

Honest Limitations

What the system CANNOT do — important to understand before building:

Limitation	Why It Exists	Impact
Cannot detect sarcasm or irony	Pattern matching checks words, not intent. "Oh, GREAT idea" sounds confident to the system	Speech score may be inflated when speaker is being sarcastic
Cannot handle multiple speakers	One camera, one mic, one face expected. Second person in frame confuses face tracking	System is designed for solo practice only
Accent-dependent STT accuracy	Speech-to-Text models are trained mostly on American/British English. Heavy accents get more transcription errors	Filler count may be inaccurate — STT might mishear words or miss fillers entirely
Lighting affects face detection	MediaPipe needs reasonable lighting to find 468 landmarks. Very dark or backlit rooms cause tracking failures	Face score becomes unreliable in poor lighting — system should warn the user
Cannot read body language below the neck	Webcam typically shows head and shoulders only	Crossed arms, fidgeting, pacing — all invisible to the system
Cultural bias in "confidence" definition	The system's definition of "confident" is Western-centric: direct eye contact = good, steady voice = good	Users from cultures where eye contact is inappropriate may score lower unfairly
Cannot distinguish thinking pauses from nervous pauses	A 3-second silence could be thoughtful or terrified — the	Silence ratio penalizes thoughtful speakers and fast

	system can't tell the difference	speakers equally

These aren't bugs to fix later. They are fundamental limitations of a browser-based, single-camera system. The right response is to be transparent about them in the UI.

What the Confidence Detector is NOT

- It is NOT a lie detector (it measures presentation STYLE, not truth)
- It is NOT a therapist (it doesn't diagnose anxiety disorders)
- It is NOT always right (a forced smile ≠ confidence, the system can be fooled)
- It is NOT surveillance (the user chooses to use it, on themselves)
- It does NOT store video unless the user explicitly asks

Why This Project Matters for Your Learning

Building the Confidence Detector will teach you:

Skill	What You'll Learn	Where It's Used Beyond This Project
Real-time media capture	Camera, microphone, streams, tracks	Video chat, streaming, security systems
Speech-to-Text	Converting audio to text in real-time	Voice assistants, transcription services, accessibility
Computer Vision	Face landmarks, expression detection	Healthcare (pain detection), gaming (avatars), security
NLP	Text pattern analysis, language scoring	Chatbots, content moderation, writing assistants
Audio Analysis	Pitch, volume, frequency processing	Music apps, hearing aids, voice authentication
AI API Integration	Multimodal AI calls, structured responses	Any app using GPT/Claude/Gemini
Real-time UI	Live dashboards, animations, data visualization	Trading platforms, monitoring systems, games
System Design	Orchestrating multiple engines, scoring algorithms	Any complex application

By the end, you won't just have a cool demo. You'll have built a **multi-input, real-time AI system** — the same architecture pattern used in self-driving cars, smart home systems, and video game AI.

Quick Summary

Question	Answer
What is it?	AI-powered presentation coaching system
What does it detect?	Facial expressions, filler words, hedging, pace,

	voice tone, eye contact
Who uses it?	Anyone preparing a presentation — professionals, students, trainers
Who decides what to improve?	The user (system measures, user decides)
What AI is used?	MediaPipe (face), Speech-to-Text + NLP, Web Audio analysis, Claude API (optional)
Does it need internet?	Face and voice analysis can run locally in the browser. If the MVP uses Web Speech API for STT, browser/network behavior depends on the browser. Fully offline STT needs an offline engine such as Vosk. Claude API always needs internet.
Platform?	Browser-first MVP. The same ideas can later be ported to desktop/mobile, but these docs assume a browser app.

Confidence Detector — System Architecture

What is System Architecture?

System architecture is the **blueprint** of how all the parts connect. Before writing code, you need to understand the big picture — what are the pieces, how do they talk to each other, and what data flows where.

The Complete System Diagram

```
+================================================================+
|                        USER'S DEVICE                           |
|                (Browser / Mobile / Desktop)                    |
|                                                                |
|  +-----------+    +------------+    +-----------+              |
|  | WEBCAM    |    | MICROPHONE |    | TOPIC     |              |
|  | (Video)   |    | (Audio)    |    | INPUT     |              |
|  +-----+-----+    +-----+------+    +-----------+              |
|        |                |                                      |
|        v                v                                      |
|  +-----------+    +------------+                               |
|  | Video     |    | Audio      |                               |
|  | Stream    |    | Stream     |                               |
|  | (30 FPS)  |    | (continuous)|                               |
|  +-----+-----+    +--+------+--+                               |
|        |             |      |                                  |
|        v             v      v                                  |
|  +-----------+  +------+ +--------+                            |
|  | FACE      |  | STT  | | VOICE  |                            |
|  | ENGINE    |  |ENGINE| | ENGINE |                            |
|  |           |  |      | |        |                            |
|  | MediaPipe |  |Speech| | Audio  |                            |
|  | FaceMesh  |  |to    | | Freq.  |                            |
|  | 468 pts   |  |Text  | | Anlys  |                            |
|  +-----+-----+  +--+---+ +----+---+                            |
```

```
|        |           |         |                                  |
|        v           v         v                                  |
|  +------------+  +-------+  +---------+                         |
|  | Expression|  | NLP   |  | Voice   |                          |
|  | Score      |  |ENGINE|  | Score   |                          |
|  |            |  |       |  |         |                         |
|  | Nervous?  |  |Filler|  | Shaky? |                           |
|  | Eye cntct?|  |Hedge |  | Pace?  |                           |
|  | Blink rt? |  |Repeat|  | Mono?  |                           |
|  +-----+------+  +---+---+  +---+-----+                         |
|        |           |          |                                |
|        +-----+-----+----+----+                                 |
|              |           |                                     |
|              v           v                                     |
|  +--------------------+ +------------------+                    |
|  | CONFIDENCE SCORING | | AI ENGINE        |                    |
|  | ALGORITHM          | | (Claude API)     |                    |
|  |                    | |                  |                    |
|  | Face  × 0.25      | | Frame + Text     |                    |
|  | Eyes  × 0.30      | | → Deep analysis  |                    |
|  | Speech× 0.30      | | Every 30 seconds |                    |
|  | Voice × 0.15      | |                  |                    |
|  | = Score 0-100     | | (OPTIONAL - v2)  |                    |
|  +--------+-----------+ +--------+---------+                    |
|           |                      |                             |
|           v                      v                             |
|  +-------------------------------------------------+           |
|  |              REAL-TIME DASHBOARD                 |           |
|  |                                                  |           |
|  |  [Webcam Feed]  [Confidence Meter]  [Alerts]    |           |
|  |  [Transcript]   [Metrics Cards]     [Timer]     |           |
|  +-------------------------------------------------+           |
|           |                                                    |
|           v  (user clicks STOP)                                |
|  +-------------------------------------------------+           |
|  |              SESSION REPORT                      |           |
|  |                                                  |           |
|  |  Final Score + Breakdown + Timeline + Coaching  |           |
|  +-------------------------------------------------+           |
+=================================================================+
```

Note: The ASCII diagram above is illustrative. For the browser-first MVP, use the scoring split defined below: **Face 0.40, Speech 0.35, Voice 0.25**.

The 5 Layers

Like ExamGuard's 5-layer architecture, our system has layers. Each layer has a clear job:

Layer 1: Input Layer (Camera + Mic)

Job: Capture raw video frames and audio data

Input	Format	Rate
Camera	Video frames (images)	30 frames per second
Microphone	Audio samples (sound)	44,100 samples per second

This layer is platform-dependent. Everything above it is platform-independent.

Layer 2: Processing Layer (4 Engines)

Job: Extract meaningful signals from raw data

Engine	Input	Output	Speed
Face Engine	Video frames	Expression, eye contact, blink rate	Real-time (every frame)
STT Engine	Audio	Text transcript	Real-time (interim + final)
NLP Engine	Transcript text	Fillers, hedging, pace, repetitions	Every final sentence
Voice Engine	Audio	Volume, pitch, steadiness	Continuous

Each engine runs independently. If one fails, the others keep working. If the mic breaks, you still get face analysis.

Layer 3: Scoring Layer

Job: Combine all engine outputs into one confidence score

```
Face Score    = expression + eye contact + blink/stress signals
Speech Score  = transcript-based signals from STT + NLP
Voice Score   = pitch + volume + steadiness signals

Confidence Score = (Face × 0.40) + (Speech × 0.35) + (Voice × 0.25)
                 = 0-100
```

These are MVP starter weights, not scientific constants. Tune them after testing with real sessions. The important design choice is that STT feeds the speech score; it is not a separate user-facing score.

WHY These Weights?

Why 0.40 / 0.35 / 0.25 and not equal thirds?

Score	Weight	Reasoning
Face	**0.40**	Mehrabian's communication research suggests ~55% of perceived confidence comes from visual cues (facial expression, eye contact, body language). Since we only see the face (no body), 0.40 is a conservative discount from 0.55. Face is also the most reliable signal — MediaPipe runs at 30 FPS with high accuracy.
Speech	**0.35**	What you SAY is what the audience remembers after the

		presentation. Filler words, hedging, and pacing are the most actionable feedback — a speaker can immediately reduce "um" count. Speech gets strong weight because it's the most improvable signal.
Voice	**0.25**	Voice tone matters, but it's the hardest to measure accurately in a browser. Web Audio API gives us volume and basic pitch, but distinguishing "nervous shaky" from "passionate emphasis" is unreliable. Lower weight = lower damage from measurement noise.

These are starting points. After testing with 10-20 real practice sessions, adjust based on what feels right. If voice analysis turns out to be more accurate than expected, bump it up.

Scoring Walkthrough with Real Numbers

Quick example to show the math in action:

```
Face Score   = 72/100  (good eye contact, slight tension detected)
Speech Score = 58/100  (8 fillers in 3 minutes, some hedging)
Voice Score  = 81/100  (steady volume, natural pitch variation)

Confidence = (72 × 0.40) + (58 × 0.35) + (81 × 0.25)
           = 28.8 + 20.3 + 20.25
           = 69.35 → rounds to 69/100

Verdict: "Good start — your voice is strong but filler words are dragging you
down"
```

For the detailed breakdown of how each sub-score is calculated, see **04_Scoring_Explained.md**.

Why Exponential Moving Average?

The confidence score updates every 1-2 seconds, but we don't just show the latest calculation. We use an **Exponential Moving Average (EMA)** to smooth it.

Think of it like a weather forecast: if Monday is 20C and Tuesday is 35C, you don't say "the weather doubled!" — you say "it's warming up." EMA does the same thing. It gives more weight to recent values but doesn't ignore the trend.

```
smoothed_score = (alpha × new_score) + ((1 - alpha) × previous_smoothed_score)

alpha = 0.3 means: 30% new value, 70% previous trend
```

Without smoothing, the meter would jump from 80 to 45 to 72 to 60 every second — that's distracting and useless. With EMA, it moves gradually: 80 → 69 → 70 → 67 — you can see the real trend.

For the full math and alpha tuning, see **04_Scoring_Explained.md**.

Layer 4: Feedback Layer

Job: Show results to the user in real-time

Component	What It Shows	Update Rate
Confidence Meter	Animated score 0-100	Every 1-2 seconds
Coaching Alerts	"Slow down", "Make eye contact"	Max 1 per 15 seconds
Live Transcript	Words appearing with highlighted fillers	Real-time (interim)
Metrics Cards	Individual scores per engine	Every 2 seconds

Layer 5: Report Layer

Job: Generate a comprehensive post-session summary

Report Section	Content
Final Score	Overall + per-engine breakdown
Timeline	Confidence score over time (chart)
Filler Analysis	Count, rate, which fillers, when
Eye Contact	% of time looking at camera
Pace Graph	WPM over time — steady or erratic
Top 3 Improvements	Rule-based coaching suggestions
AI Coaching (v2)	Claude API detailed written feedback

Data Flow: One Second of Presenting

What happens in ONE SECOND while you're presenting:

```
Second 14 of your presentation:

Camera sends 30 frames:
  → Face Engine processes each frame
  → Frame 420: Expression=neutral, EyeContact=YES, Blinks=0
  → Frame 421: Expression=neutral, EyeContact=YES, Blinks=0
  → Frame 422: Expression=tense, EyeContact=NO, Blinks=1
  → ... (30 frames total)
  → Average this second: Expression=mostly_neutral, EyeContact=87%,
BlinkRate=normal

Microphone sends audio:
  → STT gives interim: "and our revenue grew by"
  → Voice Engine: Volume=normal, Pitch=steady, NoShake
  → No final result yet (still mid-sentence)
```

```
Scoring Algorithm:
  → Face: 75/100 (mostly neutral, good eye contact)
  → Speech: waiting for sentence to finish
  → Voice: 80/100 (steady voice, normal volume)
  → Combined: ~77/100 (confident)

Dashboard updates:
  → Confidence meter shows 77
  → Color: light green
  → No coaching alert (score is good)
  → Transcript shows "and our revenue grew by" (interim, still typing)
```

What Happens When Things Fail?

Real users will deny camera access, have bad microphones, or present in dark rooms. The system must handle every failure gracefully — never crash, never show a blank screen.

Failure	What Happens	User Sees	Scoring Impact
Camera denied	Face engine cannot start	Clear message: "Camera access needed for face analysis. Grant permission or continue with voice-only mode."	Voice + Speech only (re-weight to Speech 0.58, Voice 0.42)
Microphone denied	STT and Voice engines cannot start	Clear message: "Microphone access needed for speech analysis. Grant permission or continue with face-only mode."	Face only (score = Face Score directly)
Both denied	No engines can start	"This app needs camera or microphone access to work. Please grant at least one."	Cannot score — session blocked
Face not detected	Camera works but MediaPipe can't find a face (bad lighting, too far away, face covered)	Warning: "Face not detected — check lighting and camera angle"	Fall back to Voice + Speech only scoring
STT timeout	Speech-to-Text stops returning results for >10 seconds	Auto-restart STT engine silently. If it fails 3 times, show: "Speech recognition having trouble — check microphone"	Speech score freezes at last known value until STT recovers
No audio detected	Microphone works but user isn't speaking	After 15 seconds: "Are you still presenting?	Voice score penalizes for silence ratio. Face

	(long silence)	No audio detected."	score still updates.
Browser tab hidden	User switches to another tab during session	Pause session. Show: "Session paused — camera stops when tab is hidden"	Score pauses. Resume when tab is visible again.
Low FPS / slow device	MediaPipe can't keep up at 30 FPS	Reduce to 15 FPS, then 10 FPS. Below 10: warn user	Accuracy slightly reduced but system keeps running

Design principle: Degrade gracefully, never crash. Something is always better than nothing. A voice-only confidence score is still useful.

Comparison with ExamGuard Architecture

Aspect	ExamGuard	Confidence Detector
Input	Video only	Video + Audio
Processing	Server-side (Python)	Client-side (in the app)
# of engines	1 combined detector	4 independent engines
Output	"Cheating? Yes/No + score"	"Confidence score 0-100 + coaching"
Users	Invigilators watching students	Individuals coaching themselves
Real-time feedback	Alert dashboard	Live confidence meter + coaching nudges
Post-session	Evidence screenshots + timeline	Detailed report + improvement suggestions
Camera count	Multiple cameras, multiple students	One camera, one user
Complexity	YOLO + crop + MediaPipe pipeline	MediaPipe + STT + Audio + NLP

Quick Summary

Question	Answer
How many layers?	5: Input → Processing → Scoring → Feedback → Report
How many engines?	4 processing engines: Face, Speech-to-Text, NLP, Voice
How many scored outputs?	3: Face, Speech, Voice
What's platform-dependent?	Only Layer 1 (camera/mic access). Everything else is the same
What's the hero output?	Confidence Score 0-100, updated every 1-2 seconds

How do engines connect?	They run independently, scoring layer combines their outputs
What if one engine fails?	Others keep working — degraded but functional

Confidence Detector — Detection Map

The 4 Questions to Choose Any Detection Approach

Before picking a technology for each sub-problem, ask:

```
1. WHAT signal am I trying to detect?          → Break confidence into measurable parts
2. WHAT input data do I need?                  → Video frames? Audio? Text?
3. WHAT technology can process this input?      → Pre-trained model? API? Custom code?
4. WHY this specific approach?                 → Speed? Accuracy? Cost? Offline?
```

Confidence Detection Sub-Problems

The big problem "detect confidence in a presentation" breaks down into **6 sub-problems:**

```
1. Read facial expressions     → Is the presenter tense, neutral, or relaxed?
2. Track eye contact           → Are they looking at the camera/audience?
3. Count filler words          → How many "um", "uh", "like" per minute?
4. Detect hedging language     → Are they saying "I think maybe" or stating facts?
5. Measure speaking pace       → Too fast, too slow, or steady?
6. Analyze voice steadiness    → Is the voice shaky, monotone, or confident?
```

Each sub-problem needs its own detection approach. Some share technology.

The Full Detection Map

#	Sub-Problem	Input	Technology	Why This Approach	Runs
1	**Facial Expressions**	Video frames	MediaPipe FaceMesh (468 landmarks)	Pre-trained, free, runs on any platform, fast (30+ FPS). Same tool used in ExamGuard.	Every frame
2	**Eye Contact**	Video frames	MediaPipe FaceMesh (iris + head landmarks)	Same model as #1 — no extra cost. You already built this in ExamGuard (gaze detection).	Every frame

3	**Filler Words**	Transcript text	Speech-to-Text → Pattern matching	STT converts audio to text. Pattern matching is instant and free — just check against a word list.	Every final sentence
4	**Hedging Language**	Transcript text	Pattern matching (phrase list)	Same approach as #3. Check text against known hedging phrases. No AI needed.	Every final sentence
5	**Speaking Pace**	Transcript text + time	Word count ÷ elapsed time	Simple math — no model needed. Count words, divide by seconds, multiply by 60 = WPM.	Every 5 seconds
6	**Voice Steadiness**	Raw audio	Audio frequency analysis (FFT)	Built into every platform's audio API. Measures pitch variation and volume over time.	Continuous

Optional (v2):

#	Sub-Problem	Input	Technology	Why	Runs
7	**Deep AI Analysis**	Video frame + transcript	Claude API (multimodal)	Combines face + text for nuanced coaching that rules can't provide. Costs money.	Every 30 seconds

Detailed Breakdown: Why Each Approach?

1. Facial Expressions: MediaPipe FaceMesh

Why not a dedicated emotion classifier (like FER, AffectNet)?

Option	Pros	Cons	Verdict
MediaPipe FaceMesh	Free, fast, 468 points, you know it from ExamGuard, runs offline	Doesn't label emotions directly — YOU write the distance-to-emotion logic	**CHOSEN** — gives us raw data, we control the interpretation
Dedicated emotion model (FER)	Labels emotions directly ("happy", "sad", "angry")	Often inaccurate on subtle expressions, black box (can't tune it), many are research-only	Skip — too inaccurate for subtle confidence signals
Claude API per frame	Most accurate expression reading possible	Way too slow (2-5 sec/frame) and expensive for real-time	Use periodically, not per-frame

Key decision: We use MediaPipe for RAW MEASUREMENTS and write our OWN expression logic. This gives us control — we can tune what "nervous" means rather than relying on a generic emotion classifier.

2. Eye Contact: Same MediaPipe Model

No extra technology needed — eye contact uses the SAME 468 landmarks from #1.

```
From ExamGuard you already know:
  - Iris landmarks (468-472) → where the iris is pointing
  - Head direction landmarks (4, 133, 362) → where the head is facing
  - Combine both → "looking at camera" or "looking away"
```

This is a direct reuse of ExamGuard skills. Zero new learning needed.

3 & 4. Filler Words + Hedging: Speech-to-Text → Pattern Matching

Why not use AI for filler detection?

Approach	Speed	Cost	Accuracy	Verdict
Pattern matching (word list)	Instant	Free	~85% (misses context like "I like pizza")	**CHOSEN for v1** — fast, free, good enough
Claude API per sentence	2-5 seconds	~$0.001/sentence	~98% (understands context)	v2 enhancement — add as optional layer

For v1, catching 85% of fillers instantly is better than catching 98% with a 3-second delay.

5. Speaking Pace: Simple Math

No AI needed at all. Pure arithmetic:

```
words_spoken = count words in transcript
time_elapsed = current time - session start time
pace = (words_spoken / time_elapsed) * 60  → words per minute
```

Check against thresholds: <100 = too slow, 130-160 = optimal, >180 = too fast.

6. Voice Steadiness: Audio Frequency Analysis

Every platform has a built-in way to analyze audio frequencies:

What We Measure	How	What It Tells Us
Volume over time	Measure amplitude of audio signal	Drops = trailing off. Spikes = emphasis.
Pitch variation	Measure fundamental frequency (FFT)	Monotone = low variation. Shaky = rapid small variation. Confident = natural variation.
Silence ratio	Detect when volume < threshold	Long silences = lost for words. Short pauses = thoughtful.

How Each Detection Connects to the Confidence Score

```
Detection 1: Expressions  ──→ } Face Score (0-100) ───────┐
Detection 2: Eye Contact  ──→ }                            ├──→ CONFIDENCE
Detection 3: Filler Words ──→ }                            │     SCORE
Detection 4: Hedging      ──→ } Speech Score (0-100) ──────┤     (0-100)
Detection 5: Pace         ──→ }                            │
Detection 6: Voice Steady ──→ Voice Score (0-100) ─────────┘

Weights:
  Face        × 0.40
  Speech      × 0.35
  Voice       × 0.25
  ──────────────────
  Total       = 1.00
```

These are starting heuristics for the MVP, not universal truth. Eye contact still matters a lot, but in implementation it is folded into the Face Score rather than shown as a separate top-level score.

The Blendshapes We Use

MediaPipe FaceMesh provides 52 blendshape coefficients. We don't use all 52 — most are for avatar animation (puffed cheeks, tongue out). Here are the ~15 that matter for confidence detection:

Blendshape	What It Measures	What It Tells Us About Confidence	Threshold
eyeBlinkLeft	Left eye closure (0.0 open - 1.0 closed)	Rapid blinking = stress/nervousness	>0.5 = blink detected
eyeBlinkRight	Right eye closure	Same as left — we	>0.5 = blink detected

		average both eyes	
eyeLookUpLeft	Left eye looking up	Combined with head pose = looking away from camera	>0.3 = looking up
eyeLookUpRight	Right eye looking up	Same — averaged with left	>0.3 = looking up
eyeLookDownLeft	Left eye looking down	Reading notes? Looking at desk?	>0.3 = looking down
eyeLookDownRight	Right eye looking down	Same — averaged with left	>0.3 = looking down
eyeLookInLeft	Left eye looking inward (toward nose)	Cross-referencing with right = gaze direction	>0.3 = looking right
eyeLookInRight	Right eye looking inward	Combined = looking left	>0.3 = looking left
eyeSquintLeft	Left eye squinting	Tension indicator — squinting under stress	>0.4 = squint detected
eyeSquintRight	Right eye squinting	Same	>0.4 = squint detected
browDownLeft	Left brow lowered	Furrowed brows = tension, concentration, or frustration	>0.3 = brow tension
browDownRight	Right brow lowered	Same	>0.3 = brow tension
browInnerUp	Inner brow raised	Worry/concern expression	>0.3 = worry signal
mouthSmileLeft	Left corner of mouth raised	Genuine smile (both sides) vs forced (one side)	>0.4 = smile detected
mouthSmileRight	Right corner of mouth raised	Symmetry check: genuine smiles are symmetrical	>0.4 = smile detected
jawOpen	Jaw opening amount	Speaking vs not speaking detection	>0.1 = mouth open (talking)

Why these 15 and not all 52? The remaining blendshapes cover tongue position, cheek puff, lip pucker, etc. — things that don't indicate confidence or nervousness. Using fewer signals means less noise in our scoring.

Concrete Formulas

No code here — just the math so you understand exactly what the system computes.

Eye Aspect Ratio (EAR) for Blink Detection

EAR measures how "open" an eye is. When you blink, EAR drops to near zero.

```
EAR = (|p2 - p6| + |p3 - p5|) / (2 × |p1 - p4|)

Where:
```

```
  p1, p4 = outer and inner corner of the eye (horizontal)
  p2, p3 = upper eyelid landmarks (vertical)
  p5, p6 = lower eyelid landmarks (vertical)
  |a - b| = Euclidean distance between points a and b
```

Worked example:

```
Eye wide open:
  |p2 - p6| = 12 pixels    (upper-lower distance, left side)
  |p3 - p5| = 11 pixels    (upper-lower distance, right side)
  |p1 - p4| = 30 pixels    (corner-to-corner distance)

  EAR = (12 + 11) / (2 × 30) = 23 / 60 = 0.383

Eye during blink:
  |p2 - p6| = 2 pixels
  |p3 - p5| = 1 pixel
  |p1 - p4| = 30 pixels

  EAR = (2 + 1) / (2 × 30) = 3 / 60 = 0.05

Threshold: EAR < 0.20 for 2+ consecutive frames = one blink
Normal blink rate: 15-20 per minute
Stress blink rate: >25 per minute
```

WPM Calculation (Words Per Minute)

```
WPM = (total_words / elapsed_seconds) × 60

Windowed version (last 15 seconds):
WPM_recent = (words_in_last_15_seconds / 15) × 60
```

Worked example:

```
User has spoken 47 words in the last 15 seconds:
WPM_recent = (47 / 15) × 60 = 188 WPM

Verdict: Too fast (optimal range is 130-160 WPM)
Coaching nudge: "Slow down a bit"
```

Filler Rate

```
filler_rate = filler_count / total_words × 100    (as percentage)

OR

fillers_per_minute = filler_count / elapsed_minutes
```

Worked example:

```
Transcript: "So um we decided to uh launch the product because um like
             the market was uh ready and you know we had the um resources"

Fillers found: "um" (×3), "uh" (×2), "like" (×1), "you know" (×1) = 7 fillers
Total words: 25
Elapsed time: 12 seconds

filler_rate = 7 / 25 × 100 = 28% of words are fillers (very high)
fillers_per_minute = 7 / (12/60) = 35 fillers/minute (extremely high)

Normal: <2 per minute
```

```
Acceptable: 2-5 per minute
High: 5-10 per minute → coaching nudge
Very high: >10 per minute → strong warning
```

Threshold Values and WHY

Every magic number in the system, what it is, and why that specific value:

Metric	Threshold	Value	WHY This Number
EAR blink	Blink detected	<0.20 for 2+ frames	Research standard (Soukupova & Cech 2016). Below 0.20 the eye is effectively closed. Requiring 2 frames prevents false positives from motion blur.
Blink rate	Normal	15-20/min	Medical average for adults during conversation. Below 10 = staring. Above 25 = stress indicator.
Blink rate	Stress signal	>25/min	Studies show anxious speakers blink 26-30 times per minute vs 17 average.
Eye contact	Good	>60% of time looking at camera	Presentation coaches recommend 60-70%. Below 50% feels evasive. Above 90% feels intense/staring.
WPM	Too slow	<100	Below 100 WPM sounds hesitant or under-prepared. TED talks average 130-170 WPM.
WPM	Optimal	130-160	The "sweet spot" from public speaking research. Fast enough to hold attention, slow enough to be understood.
WPM	Too fast	>180	Above 180 WPM, audiences lose comprehension. Nervous speakers often hit 200+.
Filler rate	Acceptable	<5/min	Most audiences don't consciously notice fillers below 3-5 per

			minute.
Filler rate	High	>10/min	Above 10/min, fillers become distracting. Audiences start counting them.
Hedging phrases	Flag threshold	>3 per minute	Occasional hedging is natural ("I think" once is fine). Constant hedging ("I sort of maybe think") signals low confidence.
Volume drop	Trailing off	>40% drop from speaker's average	A 40% volume drop within a sentence almost always means trailing off or losing confidence in the statement.
Pitch variation	Monotone	StdDev <10 Hz over 30 seconds	Very low pitch variation = monotone delivery. Confident speakers vary pitch by 20-50 Hz naturally.
Pitch variation	Shaky	StdDev >80 Hz over 5 seconds	Rapid, large pitch swings in a short window = voice tremor, usually from nervousness.
Silence	Thinking pause	1-3 seconds	Natural pauses for thought. Don't penalize.
Silence	Awkward gap	>5 seconds	More than 5 seconds of silence feels uncomfortable. May indicate the speaker lost their place.
Smile symmetry	Genuine	Left-right difference <0.15	Duchenne smiles (genuine) are symmetrical. Forced/nervous smiles show asymmetry >0.15 between left and right mouth corners.
Coaching alert cooldown	Min gap between alerts	15 seconds	Showing alerts more often than every 15 seconds is distracting and counterproductive. Let the speaker

			absorb one tip before giving the next.
EMA alpha	Smoothing factor	0.3	0.3 = 30% weight on new value, 70% on trend. Lower = smoother but more lag. Higher = more responsive but jumpier. 0.3 is a good balance for 1-2 second update cycles.

These are all tunable. After testing with real sessions, some will feel too aggressive (too many warnings) or too lenient (not catching real issues). Adjust based on user feedback.

Fairness and Bias

An honest discussion. This system has bias — all systems do. Here's what we know:

Face Shape and Skin Tone

Concern	Status	What We Know
Does MediaPipe work for all skin tones?	Mostly yes, with caveats	Google tested FaceMesh across skin tones and it performs well. However, very dark skin in low lighting has higher landmark error rates. This is a known issue in computer vision.
Does it work for all face shapes?	Yes for detection, maybe not for scoring	MediaPipe finds landmarks on any face shape. But our THRESHOLDS (e.g., smile symmetry, brow tension) were calibrated on limited face types. A naturally asymmetric face might trigger false "forced smile" signals.
Glasses, beards, hijab?	Partial support	Glasses: works but may interfere with eye tracking accuracy. Thick beards: jaw landmarks less reliable. Hijab/headscarf: works fine for face landmarks — FaceMesh only needs the face visible.

Accent and Language

Concern	Status	What We Know
Does STT work for all accents?	No — accuracy varies	Web Speech API (browser STT)

	significantly	is most accurate for American and British English. Indian English, Nigerian English, Southern US accents — all get more transcription errors. This directly affects filler word detection accuracy.
Non-English speakers?	Limited	Filler words are language-specific. "Um" in English is "euh" in French, "eto" in Japanese. Our word list is English-only in v1.
Speaking style differences	Not accounted for	Some cultures speak faster or slower by default. Our 130-160 WPM "optimal" range is based on Western public speaking norms.

What We Should Do About It

5. **Be transparent** — show users a note: "This system was tested primarily with American English speakers. Accuracy may vary with other accents."
6. **Let users calibrate** — a 30-second calibration phase where the system learns the user's baseline (their normal pace, their typical eye contact, their default facial expression)
7. **Never present scores as absolute truth** — always frame as "relative to your baseline" not "objectively good/bad"
8. **Don't build a filler word list for languages you haven't tested** — better to say "unsupported" than to give wrong results

This isn't something to "fix in v2." Bias awareness should be part of v1 design.

Skills You Already Have (from ExamGuard)

Detection	ExamGuard Equivalent	Reuse Level
Facial landmarks	MediaPipe FaceLandmarker	**Direct reuse** — same library, same points
Eye contact	Gaze detection (iris tracking)	**Direct reuse** — same code logic
Blink rate	Talking detection (mouth cycles)	**Pattern reuse** — same cycle-counting approach
Expression analysis	Head direction + scoring	**Concept reuse** — same distance-to-score pattern

Skills You Need to Learn (NEW)

Detection	What's New	Which Phase
Speech-to-Text	Converting audio to text in	Phase 2

	real-time	
NLP pattern matching	Analyzing text for linguistic patterns	Phase 4
Audio frequency analysis	Measuring pitch, volume from raw audio	Phase 2
AI API integration	Sending multimodal data to Claude API	Phase 5
Real-time dashboard	Live updating UI with multiple data sources	Phase 5

Quick Summary

Question	Answer
How many detections?	6 core + 1 optional (AI deep analysis)
Main technology?	MediaPipe FaceMesh (face) + Speech-to-Text (voice) + Pattern Matching (text)
What's reused from ExamGuard?	Face landmarks, iris tracking, blink detection, scoring pattern
What's new?	Speech-to-text, NLP, audio analysis, AI API, real-time dashboard
Why not use AI for everything?	Too slow and expensive for real-time. Use AI periodically (v2), rules for real-time.

How the Confidence Score Actually Works

> ***This is the most important document in the project.*** *If you understand how the score is calculated, you understand the entire system. Every detection engine, every threshold, every UI element exists to feed this score.*

The Big Picture in One Sentence

We measure three things — your **face**, your **words**, and your **voice** — give each a score out of 100, then combine them with weights into one final number.

```
Final Confidence Score = (Face Score x 0.40) + (Speech Score x 0.35) + (Voice
Score x 0.25)
```

That's it. Everything else is detail about how each sub-score is calculated.

WHY These Three Scores?

Watch the Human: How Does a Real Presentation Coach Score You?

A professional coach watches a presenter and mentally tracks three separate channels:

Channel	What They Notice	How Important
Face	"Are they making eye contact? Do they look relaxed or tense? Are they smiling naturally?"	Very — it's the first thing the audience sees
Words	"Are they saying 'um' every 5 seconds? Do they sound sure of their facts? Do they finish sentences?"	Very — it's what the audience REMEMBERS
Voice	"Is their voice steady or shaky? Are they speaking at a comfortable pace? Can I hear them clearly?"	Important — but less than face and words

Our three scores match these three channels exactly.

WHY These Weights? (0.40 / 0.35 / 0.25)

Score	Weight	WHY This Weight
Face: 0.40	Highest	Body language is 55% of communication (Mehrabian's research). Eye contact is the #1 signal audiences use to judge confidence. It's also what we detect most reliably (MediaPipe blendshapes are accurate).
Speech: 0.35	Second	What you SAY is the lasting impression. Filler words and hedging are concrete, countable signals. Pattern matching gives us reliable detection. But it only works when the person is speaking (not useful during pauses).
Voice: 0.25	Lowest	Voice tone matters, but it's the HARDEST to measure accurately in a browser. Pitch detection via FFT has noise. Volume depends on mic distance. We weight it lower because our measurement confidence is lower.

Honest note: These weights are starting heuristics, NOT scientific constants. After testing with real users, we may discover that speech matters more than face for certain presentation styles. The weights should be tuned based on real data.

WHY NOT Equal Weights (0.33 / 0.33 / 0.33)?

Equal weights assume all three signals are equally reliable AND equally important. They're not:

Signal	Detection Reliability	Audience Impact
Face (blendshapes)	High — MediaPipe is well-tested	High — first impression
Speech (pattern matching)	Medium — misses context ("I like pizza" counted as filler)	High — lasting memory
Voice (FFT in browser)	Lower — mic quality varies, ambient noise	Medium — supporting signal

Giving voice equal weight to face would let a noisy microphone swing the score too much.

How Each Score is Calculated (with Real Numbers)

Face Score (0-100)

The face score starts at a **baseline of 45** and gets adjusted up or down based on what we detect:

```
Face Score = 45 (baseline)
           + Expression adjustment  (-20 to +25)
           + Eye contact adjustment (-15 to +20)
           + Blink rate adjustment  (-10 to 0)
           + Posture adjustment     (-12 to +8)
           + Fidget adjustment      (-10 to 0)
           + Hand gesture adjustment (0 to +7)
```

Expression: What Your Face is Doing

We use MediaPipe's **52 face blendshapes** — pre-computed scores that tell us exactly what each part of the face is doing. The key ones:

Blendshape	What It Measures	Threshold	Meaning
`mouthSmileLeft + mouthSmileRight`	Average smile amount	> 0.08	Smiling — confident signal
`mouthFrownLeft + mouthFrownRight`	Frown + `mouthPressLeft/Right` (lip press)	frown > 0.12 AND press > 0.08	Tense — stress signal
`browInnerUp` + `eyeWideLeft/Right`	Brow raised + eyes wide	brow > 0.25 AND wide > 0.15	Worried — anxiety signal
`jawOpen`	Mouth opening amount	> 0.02	Speaking — active presenting (positive)

WHY blendshapes instead of measuring distances ourselves?

We actually tried manual distance calculations first (measuring pixel distances between landmarks). It failed badly:

- Our "smile detection" gave negative scores when people smiled (mouth opening shifted the center point)
- "Tense" triggered on every normal resting face (lip distance threshold was too tight)
- Thresholds that worked on one person failed on another

Blendshapes solved this because MediaPipe's model was trained on millions of faces. It already knows what a smile looks like across different face shapes, sizes, and lighting conditions. We just read the pre-computed score.

Approach	Our Experience	Verdict
Manual distance math	Failed — smile gave negative scores, tense triggered on resting faces	Abandoned
MediaPipe blendshapes	Works — `mouthSmileLeft > 0.08` reliably detects smiles	Using this
Emotion classifier (FER)	Not tested — research shows 40-60% accuracy on subtle expressions	Too unreliable

Expression Score Adjustments

Expression	Adjustment	WHY
Smiling	+25	Genuine smile is the strongest single confidence signal
Speaking	+15	Actively presenting is positive (mouth moving = engaged)
Neutral	0	Default state — not good or bad
Worried	-10	Mild negative — could be concentrating
Tense	-20	Strong negative — visible stress

Eye Contact: Looking at the Camera

We use blendshape gaze directions:

```
eyeLookDownLeft/Right  → how much looking down
eyeLookUpLeft/Right    → how much looking up
eyeLookInLeft/Right    → how much looking inward
eyeLookOutLeft/Right   → how much looking outward

If the STRONGEST gaze direction score is below 0.55 → looking at camera
If above 0.55 → looking away
```

WHY 0.55 and not 0.5?

We tested with real video. At 0.4 (our first try), a person looking straight at a laptop camera was flagged as "looking down" — because laptop cameras are slightly below eye level. 0.55 gives tolerance for normal camera angles.

Eye Contact %	Score Adjustment	WHY
> 80%	+20	Excellent — strong connection with audience
60-80%	+12	Good — natural pattern (look, think, look back)
30-60%	+5	Acceptable — might be reading notes occasionally
< 15%	-15	Poor — avoiding audience, reading everything

Blink Rate

Normal: 15-20 blinks/minute. Nervous: 25+/minute. Very anxious: 35+/minute.

We detect blinks using the `eyeBlinkLeft/Right` blendshape. When the score jumps from below 0.3 to above 0.5, that's one blink.

Blink Rate	Adjustment	WHY
< 25/min	0	Normal — no penalty
25-35/min	-5	Slightly elevated — mild stress signal
> 35/min	-10	High — anxiety indicator

Posture (from Pose Landmarks)

MediaPipe Pose Landmarker detects shoulder positions.

Posture	Adjustment	Detection Method
Upright	+8	Shoulder tilt < 0.06 AND width normal
Tilted	-5	Shoulder tilt > 0.06 (one shoulder noticeably higher)
Slouching	-12	Shoulder width very small (far from camera, hunched)

Fidgeting

We track shoulder and wrist positions across frames. Movement rate above a threshold = fidgeting.

Fidget Score	Adjustment	Detection
< 25	0	Normal movement

25-50	-5	Noticeable fidgeting
> 50	-10	Excessive movement — distracting

Subtlety we handle: Moving hands to GESTURE is positive (hand_position = "raised/gesturing" gives +7). Moving hands nervously is negative (fidget > 50 gives -10). Same body part, different meaning based on HOW it moves.

Speech Score (0-100)

The speech score analyzes the TEXT from speech-to-text:

```
Speech Score = 100 (start perfect)
             - Filler penalty     (0 to -40)
             - Hedge penalty      (0 to -30)
             - Repetition penalty (0 to -15)
             - Pace penalty       (0 to -15)
```

Filler Words

What counts as a filler and WHY:

Filler	WHY It's a Filler	Can It Be a Real Word?
um, uh, uhh, ah, er, hmm	Pure verbal pauses — zero meaning in any context	No — always fillers
like	"It was, like, really good" = filler. "I like pizza" = real word	Yes — we accept ~15% false positives
basically, actually, literally	Used as verbal padding: "So basically, the thing is..."	Sometimes meaningful, usually padding
you know, I mean	Filler phrases: "You know, the data shows..."	Almost always fillers in presentations
sort of, kind of, okay so	Softening/opening phrases with no content	Usually fillers

Filler penalty formula:

```
filler_rate = total_fillers / total_words x 100

Penalty = filler_rate x 5 (capped at 40)

Examples:
  2 fillers in 100 words = 2% rate → penalty = 10
  5 fillers in 100 words = 5% rate → penalty = 25
  8 fillers in 100 words = 8% rate → penalty = 40 (max)
```

WHY cap at 40? Because fillers are NOT the only speech signal. Even with terrible filler usage, your hedging, pace, and repetitions also matter. If fillers alone could take the score to 0, the other signals would be meaningless.

Hedging Phrases

Hedge Phrase	WHY It Shows Low Confidence
"I think", "I believe", "I feel like"	Presenting opinion as uncertain instead of fact
"maybe", "probably", "perhaps"	Weak modifiers — sounds unsure
"sort of", "kind of", "a little bit"	Downplaying your own statement
"I'm not sure", "I could be wrong"	Explicitly saying you lack confidence
"sorry", "sorry but"	Apologizing for having something to say

Hedge penalty: Each hedge phrase costs 3 points (capped at 30).

WHY is the cap lower than fillers? Because some hedging is appropriate. A scientist SHOULD say "I believe the data suggests..." — that's intellectual honesty, not low confidence. We penalize quantity, not existence.

Repetitions

```
Penalty = repetition_count x 5 (capped at 15)
```

Repetitions ("the the main point") show lost train of thought. Less impactful than fillers because they're less frequent.

Pace

WPM	Penalty	WHY
130-160	0	Optimal — natural, comfortable pace
100-130 or 160-180	-5	Slightly off — could be natural variation
< 100 or > 180	-15	Problematic — hesitating or rushing

Voice Score (0-100)

The voice score analyzes the raw audio signal:

```
Voice Score = based on:
  - Volume consistency (is voice steady or fading?)
  - Pitch variation (monotone vs natural variation vs shaky)
  - Silence ratio (appropriate pauses vs lost for words)
```

WHY this is weighted lowest (0.25):

Browser-based audio analysis has limitations:

- Mic quality varies wildly between devices
- Background noise affects readings
- Browser FFT has lower precision than dedicated audio tools

- Distance from mic changes volume independent of confidence

We use voice as a **supporting signal**, not a primary one. If face and speech say "confident" but voice says "shaky," the score drops slightly — not dramatically.

Complete Walkthrough: Real Example

Scenario: A presenter gives a 2-minute practice session.

What the System Observes

Face (over 2 minutes):

- Expression: speaking 60% of time, neutral 30%, smiling 10%
- Eye contact: 75% looking at camera
- Blink rate: 22/min (normal)
- Posture: upright throughout
- Fidgeting: score 15 (minimal)
- Hand gestures: occasional (raised 20% of time)

Speech (transcript analysis):

- 280 words spoken in 2 minutes = 140 WPM (optimal)
- 6 fillers found: "um" (3), "like" (2), "basically" (1)
- Filler rate: 6/280 = 2.1%
- 2 hedging phrases: "I think", "kind of"
- 0 repetitions

Voice:

- Volume: steady, slight drop at end of sentences
- Pitch: natural variation, no shakiness
- Silence ratio: 15% (appropriate pauses)

How the Score is Calculated

Step 1: Face Score

```
Baseline:                          45
Expression (mostly speaking):     +15
Eye contact (75%):                +12
Blink rate (22/min, normal):       +0
Posture (upright):                 +8
Fidgeting (15, minimal):           +0
Hand gestures (occasional):        +3
                                 ────
Face Score:                        83
```

Step 2: Speech Score

```
Starting:                         100
Filler penalty (2.1% x 5):       -10
Hedge penalty (2 x 3):            -6
```

```
Repetition penalty (0):          +0
Pace penalty (140 WPM, optimal): +0
                                 ────
Speech Score:                    84
```

Step 3: Voice Score

```
Volume (steady):                 85
Pitch (natural variation):       80
Silence ratio (appropriate):     90
                                 ────
Voice Score (average):           85
```

Step 4: Combined Final Score

```
Final = (Face x 0.40) + (Speech x 0.35) + (Voice x 0.25)
      = (83 x 0.40)   + (84 x 0.35)     + (85 x 0.25)
      = 33.2           + 29.4             + 21.25
      = 83.85
      ≈ 84
```

Verdict: 84 = "Confident" — This presenter has good eye contact, minimal fillers, steady voice. Room to improve: reduce the 6 fillers and 2 hedges.

"What Should the Score Be?" Decision Table

Test your understanding. Read the scenario, guess the score, then check.

#	Face	Speech	Voice	Expected Score	WHY
1	Smiling, 90% eye contact, upright	0 fillers, direct statements, 145 WPM	Steady, clear	**90+**	Everything positive — highly confident
2	Neutral, 80% eye contact, upright	3 fillers (1.5%), 1 hedge, 140 WPM	Steady	**78-82**	Good overall, minor filler issue
3	Neutral, 40% eye contact, upright	0 fillers, 135 WPM	Steady	**65-70**	Good speech but poor eye contact drags face score down
4	Tense, 30% eye contact, tilted	10 fillers (5%), 5 hedges, 190 WPM	Shaky, fast	**25-35**	Multiple stress signals across all three channels
5	Smiling, 85% eye contact	8 fillers (4%), 4 hedges, 180 WPM	Slightly fast	**55-65**	Face says confident but speech says nervous —

					mixed signals
6	Neutral, 70% eye contact	No speech (silent video)	No audio	**65-73**	Face-only scoring — no speech or voice data
7	Neutral, 90% eye contact, gesturing	1 filler, 0 hedges, 150 WPM	Perfect	**85-90**	Great across all channels, gestures help
8	Worried, 20% eye contact, slouching	15 fillers (7.5%), 8 hedges, 85 WPM	Monotone, quiet	**15-25**	Everything signals low confidence
9	Speaking, 60% eye contact	4 fillers (2%), 2 hedges, 165 WPM	Slightly fast	**65-72**	Decent but room for improvement everywhere
10	Smiling, 95% eye contact, gesturing	2 fillers (0.8%), 0 hedges, 150 WPM	Confident, varied	**92-95**	Near-perfect presentation delivery

---

How to Tune the Weights

The weights (0.40 / 0.35 / 0.25) are starting values. Here's how to improve them:

Step 1: Record 5-10 Test Sessions

Record yourself presenting with DIFFERENT confidence levels:

- One session where you TRY to be confident
- One session where you deliberately add lots of "ums"
- One session where you avoid eye contact
- One session at normal comfort level

Step 2: Score Each Session by Human Judgment

Watch each recording and rate it 0-100 yourself. Be honest.

Step 3: Compare Human Score vs System Score

Session	Human Says	System Says	Gap
Confident attempt	82	78	-4 (close!)
Lots of ums	35	55	+20 (system too generous)
No eye contact	40	60	+20 (system too generous)
Normal	65	68	+3 (close!)

Step 4: Adjust Weights to Close the Gap

If the system is too generous on "lots of ums" → increase Speech weight.

If the system is too generous on "no eye contact" → increase Face weight or adjust eye contact penalty.

This is the real path to accuracy. No amount of theoretical weight-choosing beats testing with real data.

Smoothing: Why the Score Doesn't Jump Around

Without smoothing, the score would change every frame — jumping from 72 to 45 to 81 in one second. That looks broken.

Analogy: Weather Forecast

A weather app doesn't say "It's 25 degrees! Now it's 23! Now it's 26!" every minute. It smooths the data: "Today will be around 24-25 degrees."

We use **Exponential Moving Average (EMA)**:

```
New displayed score = (0.3 x latest reading) + (0.7 x previous displayed score)
```

This means:

- 70% of the score comes from where it WAS (stability)
- 30% comes from the new reading (responsiveness)

Example:

```
Frame 1: Raw score = 75 → Displayed: 75 (first reading)
Frame 2: Raw score = 60 → Displayed: (0.3 x 60) + (0.7 x 75) = 18 + 52.5 = 70.5
≈ 71
Frame 3: Raw score = 80 → Displayed: (0.3 x 80) + (0.7 x 71) = 24 + 49.7 = 73.7
≈ 74
Frame 4: Raw score = 50 → Displayed: (0.3 x 50) + (0.7 x 74) = 15 + 51.8 = 66.8
≈ 67
```

Instead of jumping 75→60→80→50, the displayed score moves 75→71→74→67. Much smoother, still responsive to real changes.

Common Scoring Mistakes to Watch For

Mistake	What Goes Wrong	How to Fix
"Score is always 50-60"	Neutral expression + decent eye contact + no speech = baseline score	This is correct! Not presenting = moderate score. Score should only be high when ACTIVELY presenting well.
"Score jumps wildly"	No smoothing applied, or smoothing factor too low	Apply EMA with 0.7 stability factor
"Smiling person gets low	Lots of fillers in speech	Check speech analysis — the

score"	override the face signal	COMBINED score is what matters
"Silent person gets high score"	Only face score active (no speech/voice data)	When speech is unavailable, show "Face only" label and don't claim overall confidence
"Score is too high for nervous person"	Weights favor face too much and person looks calm but sounds terrible	Test with deliberate filler-heavy sessions, tune Speech weight up
"Score drops when looking down briefly"	Eye contact window too short, single glance down tanks the score	Increase rolling window (we use last 30 readings — about 1 second)

Score Labels

Score Range	Label	Color	What It Means
85-100	Highly Confident	Bright Green	Excellent delivery — ready for any audience
70-84	Confident	Green	Strong presentation — minor areas to polish
50-69	Moderate	Yellow	Decent but noticeable room for improvement
25-49	Developing	Orange	Multiple confidence signals need work
0-24	Low Confidence	Red	Significant nervousness across all channels

---

Mini Summary

- Three scores:** Face (0.40 weight), Speech (0.35), Voice (0.25)
- Face uses blendshapes** — 52 pre-computed facial action scores from MediaPipe, NOT manual distance math (we tried that, it failed)
- Speech uses pattern matching** — count fillers and hedges against word lists. Fast, free, runs locally
- Voice is weighted lowest** because browser audio analysis is the least reliable
- Smoothing (EMA)** prevents the score from jumping around — 70% stability + 30% new reading
- Tune with real data** — record yourself, compare human judgment to system score, adjust weights
- These weights are starting heuristics** — not final truth. Real accuracy comes from testing.